

1. Introduction

Project Description:

This project aims to develop a system that converts natural language queries into SQL statements using a Large Language Model (LLM). The model will be trained on appropriate benchmark datasets (e.g., Spider, WikiSQL, or other suitable datasets) and integrated into a web-based application where users can input natural language queries and receive SQL outputs. The system will only generate SQL queries and will not execute them on any database.

Purpose:

The purpose of this project is to design and implement a Text-to-SQL conversion system that leverages LLM technology to facilitate the transformation of natural language queries into SQL format. By doing so, the project aims to demonstrate the applicability of LLMs in structured query generation and to provide a functional platform for testing and evaluating this capability.

2. Project Scope

- Development of a text-to-SQL conversion model using a Large Language Model.
- Training and evaluation based on selected benchmark datasets (e.g., Spider, WikiSQL, or others determined during the project).
- Implementation of a backend using Java Spring Boot to manage API requests.
- Integration of the trained LLM via a Python FastAPI service.
- Development of a web interface for user interaction, enabling natural language input and SQL output visualization.
- The system will focus solely on query conversion; it will not execute SQL queries on any live database.

3. Stakeholder Analysis

Role	Description
End Users	Users who want to obtain SQL queries from natural language statements.
Developers	Ramazan Bozkurt, Abdulkadir Sözgen
Project Advisor	Dr. Yılmaz Kemal Yüce

4. Requirements

4.1. Functional Requirements

ID	Requirement Description	Priority
FR1	The system shall accept natural language input from users.	High
FR2	The system shall convert natural language input into SQL statements.	High
FR3	The system shall support core SQL structures (SELECT, WHERE, JOIN, GROUP BY, ORDER BY).	High
FR4	The system shall display the generated SQL statements through the web interface.	High
FR5	The LLM model shall be trained and evaluated using selected benchmark datasets.	High

4.2. Non-Functional Requirements

ID	Requirement Description	Priority
NFR1	The system shall achieve the targeted accuracy level (e.g., $\geq 80\%$) on the benchmark dataset.	High
NFR2	The system shall respond to a query within a maximum of 5 seconds.	High
NFR3	The web interface shall be compatible with modern browsers.	Medium
NFR4	User queries shall not be stored persistently.	Medium

5. Acceptance Criteria

- The LLM model achieves the target accuracy level on the selected benchmark dataset.
- The system provides valid SQL outputs for given natural language queries.
- Integration between Java Spring Boot backend and Python FastAPI model service operates successfully.
- The web interface performs essential functionalities without errors.